

Summary

First-principles AI systems engineer and applied researcher who builds across software, AI, physics, and operations. Work spans production voice agents, call intelligence, AI inspection, sub-quadratic attention research, EMI shielding simulation, EV autonomy, and referral-marketplace product design. Strongest in end-to-end ownership: problem framing, backend architecture, evaluation, UI, automation, deployment, and cross-functional collaboration with product, sales, operations, and engineering stakeholders.

Professional Experience

DriveX Mobility

Feb 2025 - Present

Software Engineer, AI Systems

Bengaluru, India · Official title: Technical Associate Program Manager

- Designed and shipped **MIRA**, a production outbound voice AI agent for live sales leads across **5 Indian languages**; built the ASR/TTS/LLM orchestration, routing logic, backend services, and deployment path.
- Re-architected the response path from sequential calls to a streamed, parallel service flow; reduced first response from **7s to <500ms** across **4 architecture iterations** over 68 days while preserving CRM sync and WhatsApp handoffs.
- Shipped **AIDEN**, a full-stack call analytics platform, in **31 days**; built React dashboards, backend ingestion, speaker diarization, emotion scoring, multi-model LLM analysis, and Zoho CRM embedding.
- Deployed analytics services to Kubernetes with Celery/Redis workers, database-backed reports, browser E2E coverage, and structured evaluation across **8 SOP dimensions** for sales-quality review.
- Shipped **Veronica** for AI vehicle inspection across **1,000+ parts**, **FuryX** as a Chrome Extension plus Strapi CMS portal, and **QE automation** with JMeter, Selenium, and Playwright regression suites.

Formula Manipal

May 2022 - Jan 2024

Operations Lead

- Led autonomous path planning for the FM23e EV; track-side testing improved control accuracy by **40%** and supported a 1st-place Cost and Manufacturing result at Formula Bharat 2024 with **INR 60,00,000** in sponsorships.

Tescom

2024

Hardware Engineering Intern

- Engineered a custom PCB testing solution and optimized assembly-line logic to increase error detection accuracy and reduce operational downtime.

Research & Projects

AI, physics, and applied systems

Wave Field Attention PAPER

Proposed a sub-quadratic long-context attention mechanism that reframes token interaction as wave propagation; derived an $O(n \log n)$ formulation, formal assumptions, and benchmark plan against quadratic softmax attention.

EMI Shielding Physics Engine PAPER

Built a vectorised EMI shielding simulation engine for multi-phase composites using Schelkunoff modelling; ran **100 kHz-10 GHz** sweeps, estimated 20-year material performance, and shipped React Native tooling with Jenkins/SonarQube CI.

Quantum Options Pricing

Built Qiskit Monte Carlo pricing workflow for financial derivatives with a **70% speedup**.

Refr Referral Marketplace

Built an Expo/React Native app for verified employee referrals in India; implemented **2-sided seeker/referrer flows**, trust metrics, referral pipeline tracking, chat, and gamified earnings.

Technical Skills

AI SYSTEMS	Agentic orchestration, real-time voice pipelines, MCP, tool use, structured outputs, speaker diarization, streaming ASR, intent classification, RAG, YOLOv8, Pipecat, Qiskit.
ENGINEERING	Python, Django, FastAPI, React, TypeScript, Next.js, SQL, C++, JavaScript, Bash, React Native.
INFRA & QE	Docker, Kubernetes, Celery, Redis, PostgreSQL, AWS S3, Jenkins, SonarQube, Playwright, Selenium, JMeter, Agile.
INTEGRATIONS	Zoho CRM, VoBiz Telephony, WhatsApp Business API, Sarvam AI, Hume AI, Strapi CMS, MetaX, Chatwoot.

Education & Certifications

Manipal Institute of Technology · B.Tech in Electrical and Electronics Engineering	2025
IIT Roorkee (iHUB) · Executive PG Certificate in Full Stack Development	2025
Certifications · Python for Data Science - IBM; SQL for Data Science - Coursera; Cloud Computing - IBM.	